FX Equity Spillover

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1 Abstract

The goal of this model is to analyze and develop a strategy for trading FX using equity returns. There is a relationship between foreign equity returns and their currency. While the relationship between a country's equity returns and their corresponding currencies has been discussed, this work does contribute or focus to the economic interpretation or consequences. Instead, it investigates whether the relationship can be effectively modeled and exploited for trading. [1]

2 Historical Data

The initial model was developed using FX futures collected from Bloomberg Terminal and constructed using front month roll adjustments. Although FX futures do not fully capture the FX market, they provide a sufficient cross-sectional coverage for these models. FX Futures were selected because they provide a convenient instrument for both historical backtesting and real-world trading execution. From a backtesting perspective, FX futures are advantageous since they embed the carry accrued from holding the underlying foreign currency, simplifying data handling, especially for non-deliverable forward currencies.[2] From an implementation standpoint, FX futures are easily financeable, of highly liquid pairs, and have high quality data.

Some of the FX futures have low liquidity. As a result, the initial model includes only the most liquid contracts. The table below summarizes the liquidity status of each future. A natural step is to incorporate FX carry total returns indices. Bloomberg Terminal offers indices for 33 pairs versus USD, which include the appropriate interest rate adjustments. These illiquid futures and indices will be incorporated in subsequent extensions of the model.

MSCI ETFs were used for proxy each country's equity returns. Rather than using index levels the ETF total returns were collected and used.

[1] A GitHub repo with the full codebase and results can be found https://github.com/diegodalvarez/FXEquitySpillover
[2] See strategy ten in *Advanced Futures Trading Systems* by Rob Carver for a clear explanation of carry in FX futures.

Future	MSCI ETF	FX/Country	Liquidity	Start Date	End Date
AD	EWA	Australian Dollar	Liquid	1996-03	2026-01
BP	EWU	British Pound	Liquid	1996-03	2026-01
BR	EWZ	Brazilian Real	Liquid	2000-07	2026-01
CD	EWC	Canadian Dollar	Liquid	1996-03	2026-01
EC	EZU	Euro Foreign	Liquid	2000-08	2026-01
IS	EIS	Israeli Shekel	Illiquid	2008-03	2026-01
JY	EWJ	Japanese Yen	Liquid	1996-03	2026-01
KO	EWY	Korean Won	Illiquid	2006-09	2026-01
NO	ENOR	Norwegian Krone	Illiquid	2012-01	2026-01
NV	ENZL	New Zealand Dollar	Liquid	2010-09	2026-01
PE	EWW	Mexican Peso	Liquid	1996-03	2026-01
PP	EPOL	Polish Zloty	Illiquid	2010-05	2026-01
SE	EWD	Swedish Krona	Illiquid	2002-05	2026-01
SF	EWL	Swiss Franc	Liquid	1996-03	2026-01
SIR	INDA	Indian Ruppee	Illiquid	2013-01	2026-01
UCA	MCHI	Offshore Chinese Renminbi	Illiquid	2012-09	2026-01

3 Theoritical Approach

Following Cendese et. al [2] the theoretical foundation around this model follows a no-arbitrage constraint. For a domestic and foreign equity return R_d and R_f where the foreign return has to be scaled by some currency s_t

$$\mathbb{E}\left(R_d \cdot m_h\right) = \mathbb{E}\left(R_f \cdot \frac{s_{t+1}}{s_t} \cdot m_h\right) = 1 \tag{1}$$

m_h is the discounting factor to maintain no-arbitrage also called the stochastic discounting factor. Then rearrange the formula using covariance rules.

$$\mathbb{E}\left(R_d\right) \cdot \mathbb{E}\left(m_h\right) + \operatorname{cov}\left(m_h, R_d\right) = \mathbb{E}\left(R_f \cdot \frac{S_{t+1}}{S_t}\right) \cdot \mathbb{E}\left(m_h\right) + \operatorname{cov}\left(m_h, R_f \cdot \frac{S_{t+1}}{s_t}\right) = 1 \tag{2}$$

Then let $\mathbb{E}\left(m_h\right)$ be the one-period, risk free zero-coupon bond R_b in country d.

$$\mathbb{E}\left(R_d\right) \cdot \frac{1}{R_b} + \operatorname{cov}\left(m_h, R_d\right) = \mathbb{E}\left(R_f \cdot \frac{S_{t+1}}{S_t}\right) \cdot \frac{1}{R_b} + \operatorname{cov}\left(m_h, R_f \cdot \frac{S_{t+1}}{s_t}\right) = 1 \tag{3}$$

The equation above is useful since it separates the contribution of risk premia and the risk free rate. In a no-arbitrage setting, the stochastic discount factor m_h discounts the returns back to one. If the discounting is too slow - corresponding to positive excess returns - the covariance term $\operatorname{cov}(m_h, R_d)$ adjusts to offset these returns back to one. Conversely, if discounting is too fast, implying negative excess returns, the covariance term adjusts in the opposite direction. Then re-arrange the formula to separate the equity returns (excess returns) and the risk premia

$$\underbrace{\mathbb{E}\left(R_f \cdot \frac{S_{t+1}}{S_t}\right) \frac{1}{R_b} - \mathbb{E}\left(R_d\right) \cdot \frac{1}{R_b}}_{\text{excess equity return differential}} = \underbrace{\operatorname{cov}\left(m_h, R_d\right) - \operatorname{cov}\left(m_h, R_f \cdot \frac{S_{t+1}}{S_t}\right)}_{\text{difference in risk premia}} \tag{4}$$

Now, to fully isolate the currency returns, use a similar approach for log returns. Therefore,

$$\mathbb{E}\left(\exp\left(\log\left(m_h R_f \frac{S_{t+1}}{S_t}\right)\right)\right) = \mathbb{E}\left(\exp\left(\log\left(m_h R_d\right)\right)\right) \tag{5}$$

Equation 5 can be rewritten as

$$\mathbb{E}\left(\exp\left(\tilde{m} + r_f + \Delta s_{t+1}\right)\right) = \mathbb{E}\left(\exp\left(\tilde{m} + r_d\right)\right) \tag{6}$$

Where r_d, r_f, and Δs_{t+1} are the log domestic and foreign equity with changes in the currency, $\tilde{m}$ is $\log(m_h)$. Then with some variance and covariance rules we can get the log-comparable to equation 3.

$$\mathbb{E}(\tilde{m}) + \mathbb{E}(\tilde{r_f}) + \mathbb{E}(\Delta s_{t+1}) + \frac{1}{2}\text{var}(\tilde{m} + r_f + \Delta s_t) = \mathbb{E}(\tilde{m}) + \mathbb{E}(r_d) + \frac{1}{2}\text{var}(\tilde{m} + r_d) \tag{7}$$

Then re-arrange

$$\mathbb{E}(\tilde{r_f}) + \mathbb{E}(\Delta s_{t+1}) - \mathbb{E}(r_d) = \frac{1}{2}\left(\text{var}(\tilde{m} + r_f + \Delta s_t) - \text{var}(\tilde{m} + r_d)\right) \tag{8}$$

$$= \frac{1}{2}(\text{var}(r_f + \Delta s_t) - \text{var}(r_d)) + \text{cov}(\tilde{m}, r_f + \Delta s_{t+1}) - \text{cov}(\tilde{m}, r_d) \tag{9}$$

Which results in an equation very similar to equation 3 but in log form

$$\mathbb{E}(\Delta s_{t+1}) = \underbrace{\mathbb{E}(r_d) - \mathbb{E}(\tilde{r_f})}_{\substack{\text{expected log} \\ \text{return differential}}} + \underbrace{\text{cov}(\tilde{m}, r_f + \Delta s_{t+1}) - \text{cov}(\tilde{m}, r_d)}_{\substack{\text{difference in} \\ \text{risk premia}}} + \underbrace{\frac{1}{2}(\text{var}(r_f + \Delta s_t) - \text{var}(r_d))}_{\text{log-variance adjustment}} \tag{10}$$

The approach is appealing because it preserves flexibility in the sign of the covariance term $\text{cov}\left(m_h, R_j \frac{S_{t+1}}{S_t}\right)$. Hau and Rey [5] argue that portfolio rebalancing by international investors implies a negative relationship, whereas Bohn and Tesar[1], and Froot, Scharfstein, and Stein[4] document evidence consistent with a positive relationship.

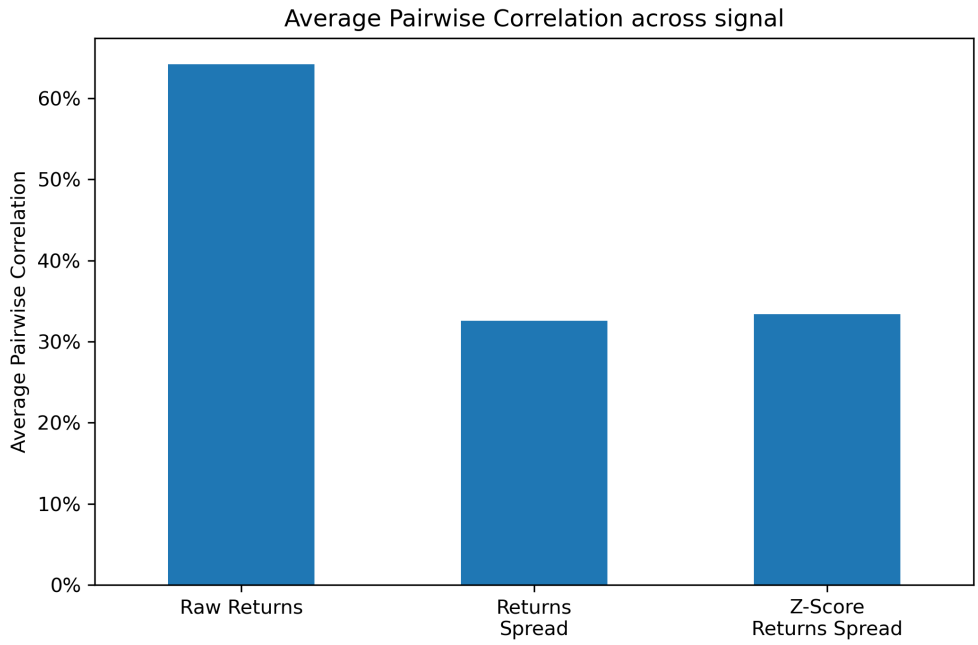
Cendese et al further not that this framework is not restricted to returns and can be extended to other equity factors such as dividend yield, value, and momentum. However, this paper (in its current status) focuses exclusively on equity returns and adopts are simpler signal construction method. Specifically, the strategy is based on the returns differential, which is then transformed using a 30-day exponentially weighted z-score.

4 Signal Analysis

The returns described in the previous section exhibit low correlation. The figure below presents the average pairwise correlation across the raw equity returns. This correlation is further reduced when considering the return spread ($R_d - R_f$) and its 30-day exponentially weighted z-score.

The return differential may be strongly correlated with the underlying equity trend. To investigate this, compute the pairwise correlations between the signal (as well as its z-score) and a generic trend-following signal. To calculate trend a simple EWMA was used as described in Bouchaud [3]. The correlation is quite low

Asset Class	Equity		FX		Equity		FX		Equity		FX	
lookback	5		5		30		30		100		100	
	Rtn Spread	30d ZScore	Rtn Spread	30d ZScore	Rtn Spread	30d ZScore	Rtn Spread	30d ZScore	Rtn Spread	30d ZScore	Rtn Spread	Rtn Spread
Country												
Australia	0.439	0.427	0.310	0.296	0.189	0.117	0.130	0.059	0.096	0.035	0.066	0.005
Brazil	0.564	0.520	0.404	0.347	0.256	0.126	0.177	0.062	0.139	0.027	0.093	-0.000
UK	0.338	0.335	0.245	0.243	0.124	0.082	0.100	0.055	0.034	0.017	0.048	0.011
Canada	0.378	0.351	0.269	0.254	0.177	0.097	0.127	0.056	0.098	0.029	0.074	0.008
Eurozone	0.410	0.369	0.333	0.295	0.160	0.063	0.137	0.056	0.050	-0.013	0.061	0.002
Japan	0.362	0.340	0.187	0.158	0.157	0.092	0.080	0.037	0.083	0.023	0.040	0.009
Mexico	0.492	0.455	0.310	0.275	0.216	0.106	0.129	0.050	0.099	0.015	0.058	0.001
New Zealand	0.400	0.366	0.269	0.239	0.134	0.064	0.122	0.052	0.035	0.004	0.061	0.002
Switzerland	0.322	0.312	0.262	0.261	0.136	0.085	0.104	0.059	0.069	0.024	0.041	0.009



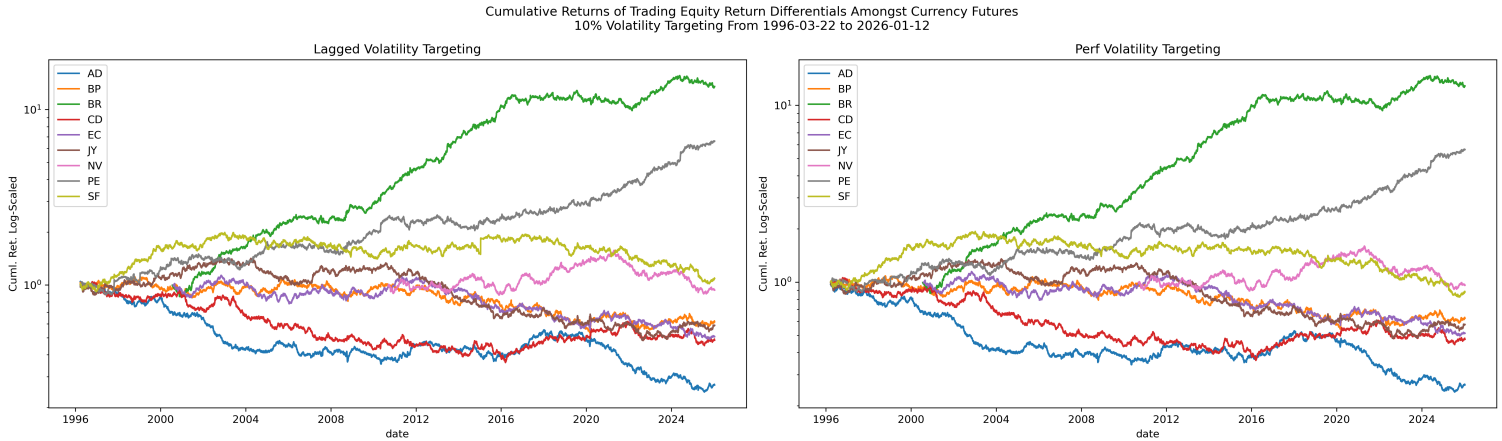
5 Trading Z-Scores

Start by computing a 30-day exponentially weighted z-score of the daily foreign–domestic equity return differential. This z-score is then used as a trading signal, which is scaled to target an annualized volatility of 10% through daily rebalancing.

Two approaches to volatility targeting are considered: a "perfect foresight" (contemporaneous) volatility scaling and a lagged volatility targeting scheme. While the lagged specification is the correct implementable approach, the higher-frequency (daily) nature of the data implies that sharp intraday volatility spikes can lead to material losses before risk adjustments take effect.

In practice, however, an investor or broker would likely adjust positions intraday in response to extreme market moves, mitigating some of this effect. Such discontinuities are more prevalent in commodities than in currencies, but the exercise remains informative as it quantifies the Sharpe ratio cost of not incorporating intraday volatility management.

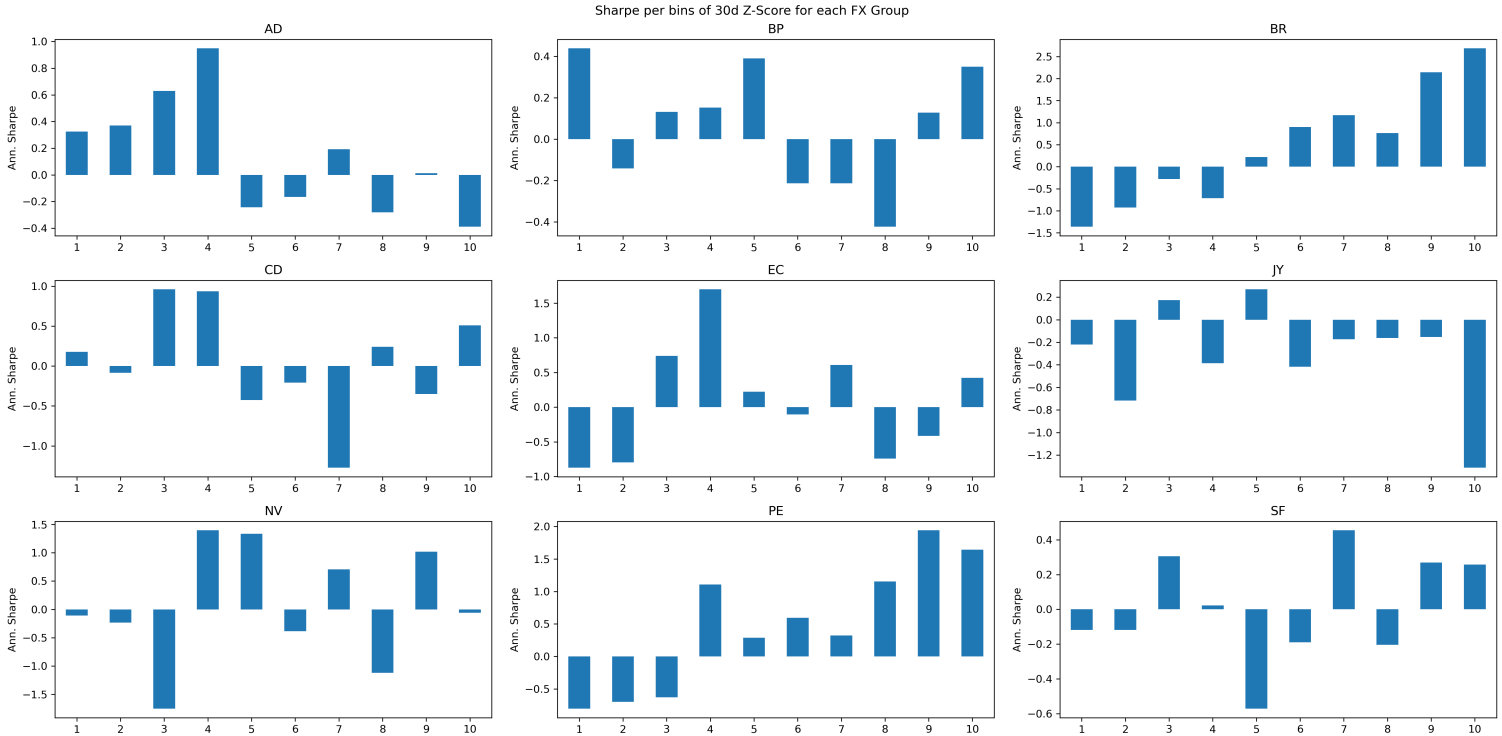
The results from trading FX futures are presented below. While the performance is not particularly strong, the returns appear to be relatively stable over time. This suggests that there may be some scope for further refinement and data-driven optimization of the signal.



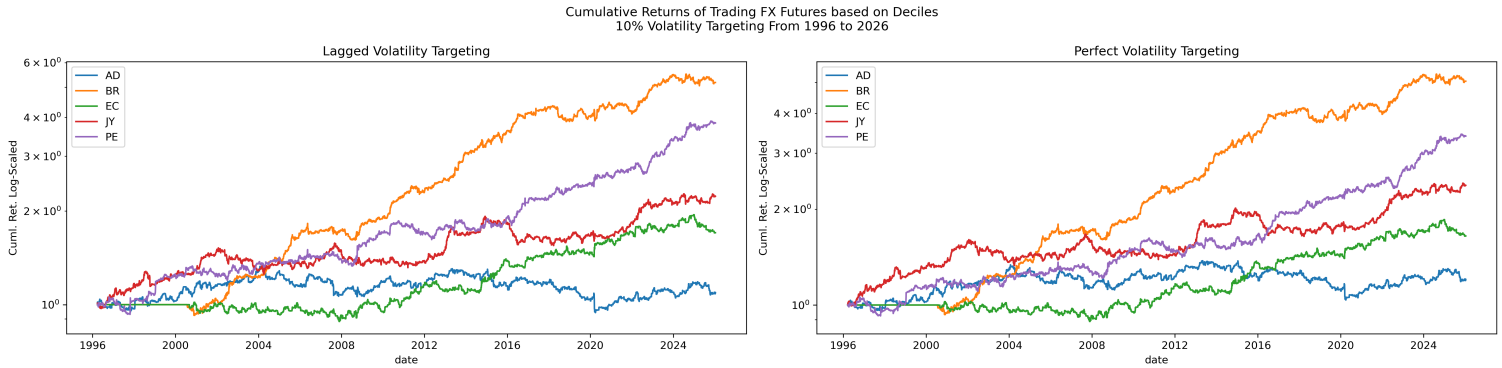
Below is a table of the annualized sharpes.

Fut Ticker	AD	BP	BR	CD	EC	JY	NV	PE	SF
Lagged Vol	-0.384	-0.110	1.074	-0.192	-0.219	-0.125	0.006	0.671	0.080
Perfect Vol	-0.413	-0.114	1.091	-0.207	-0.222	-0.142	0.021	0.654	0.005

To optimize the signal, we compute full-sample deciles of the z-score and evaluate Sharpe ratios within each bin. The results below show clear cross-sectional variation in performance. Currencies such as Brazil and Mexico exhibit momentum-like behavior (positive FX–equity correlation), whereas Australia displays the opposite pattern.



Below are the results of the optimized Z-score FX portfolio. Performance improves noticeably, suggesting that the optimization helps reduce noise, lower turnover and transaction costs, and improve signal orthogonality across strategies.



Below is a table of the annualized sharpes over the full period.

Vol Target	AD	BR	EC	JY	PE
Lagged	0.125	1.638	0.572	0.714	1.116
Perfect	0.212	1.697	0.569	0.796	1.118

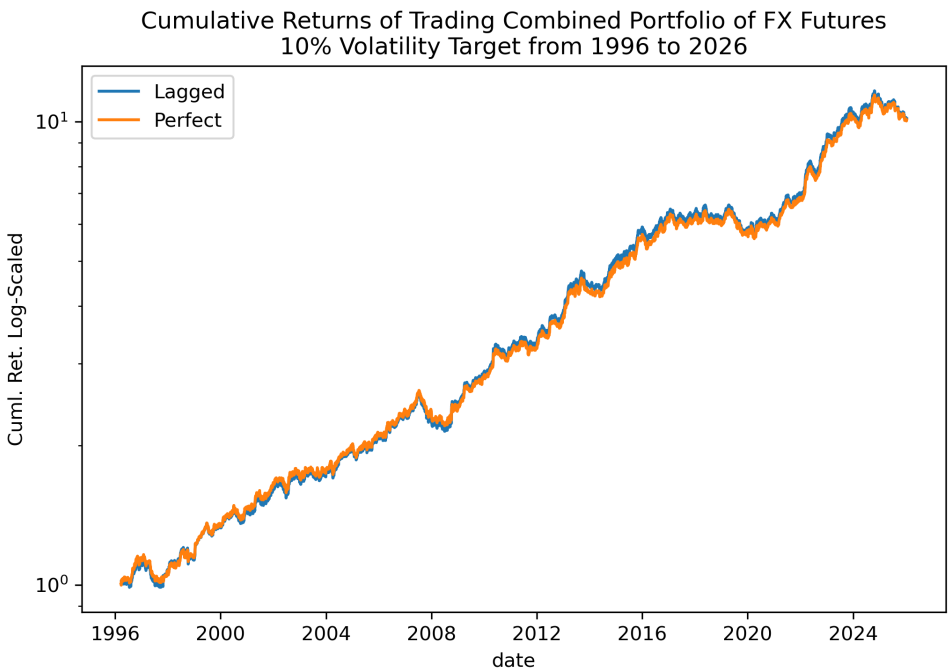
The correlation of the z-score decile optimized strategies are low, below is a correlation matrix.

Vol Target	Lagged					Perfect				
	AD	BR	EC	JY	PE	AD	BR	EC	JY	PE
AD	1.000	-0.187	-0.529	-0.156	-0.173	1.000	-0.192	-0.533	-0.151	-0.185
BR	-0.187	1.000	0.094	-0.016	0.411	-0.192	1.000	0.104	-0.018	0.416
EC	-0.529	0.094	1.000	0.296	0.154	-0.533	0.104	1.000	0.287	0.162
JY	-0.156	-0.016	0.296	1.000	0.073	-0.151	-0.018	0.287	1.000	0.077
PE	-0.173	0.411	0.154	0.073	1.000	-0.185	0.416	0.162	0.077	1.000

Sharpe ratios must be adjusted for periods with no exposure. The decile optimization residualizes the signal and trades only when optimal, resulting in a maximum time-in-market of 40%. The portfolio Sharpe is therefore scaled to reflect this reduced exposure.

Vol Target	Lagged		Perfect	
	Adj Sharpe	Raw Sharpe	Adj Sharpe	Raw Sharpe
AD	0.177	0.125	0.300	0.212
BR	1.661	1.638	1.721	1.697
EC	0.813	0.572	0.809	0.569
JY	0.718	0.714	0.801	0.796
PE	1.133	1.116	1.134	1.118

Taking equal weights, which should result in roughly equal volatility (since each strategy is volatility-targeted), yields the following portfolio.



Vol Target	Adj Sharpe	Raw Sharpe
Lagged	1.119	1.119
Perfect	1.184	1.184

6 Trading OLS Residuals

This section evaluates trading the OLS regression of the returns differential against FX futures. Instead of using the raw return differential use the 30 day exponentially weighted z-score. The table below reports the regression results: one specification uses raw FX futures returns, while the other fits the z-score to 10%

volatility-targeted FX futures returns.

$$R_f \sim \beta(R_d - R_{for}) + \alpha + \epsilon \tag{11}$$

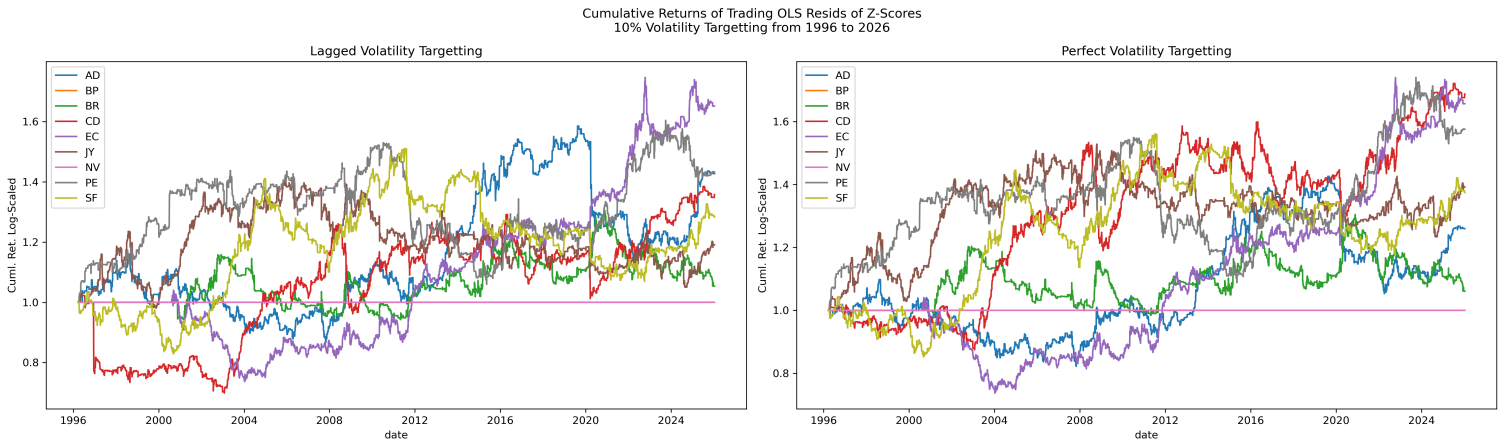
$$R_f \left(\frac{0.1}{\sigma_{t-1}\sqrt{252}} \right) \sim \beta(R_d - R_{for}) + \alpha + \epsilon \tag{12}$$

For futures returns R_f, domestic equity returns R_d, and foreign equity returns R_{for}.

		Param Val		p-value		t-stat	
	Param	α	Z-Score	α	Z-Score	α	Z-Score
Ticker	Target						
AD	No Target	0.0001	-0.0001	0.4709	0.1808	0.7210	-1.3384
	Vol Target	0.0000	-0.0001	0.7574	0.1130	0.3089	-1.5850
BP	No Target	0.0000	-0.0000	0.7450	0.8745	0.3252	-0.1579
	Vol Target	0.0000	-0.0001	0.5473	0.4500	0.6019	-0.7555
BR	No Target	0.0003	0.0008	0.0204	0.0000	2.3191	5.9876
	Vol Target	0.0001	0.0005	0.0575	0.0000	1.8999	5.7306
CD	No Target	0.0000	-0.0000	0.7744	0.7839	0.2867	-0.2742
	Vol Target	-0.0000	0.0000	0.6989	0.9897	-0.3868	0.0129
EC	No Target	0.0000	0.0000	0.6756	0.5404	0.4185	0.6123
	Vol Target	0.0000	0.0000	0.8577	0.9223	0.1793	0.0975
JY	No Target	-0.0001	-0.0001	0.0807	0.1825	-1.7467	-1.3332
	Vol Target	-0.0002	-0.0001	0.0048	0.1756	-2.8202	-1.3547
NV	No Target	0.0000	0.0001	0.8094	0.4335	0.2412	0.7833
	Vol Target	-0.0000	0.0001	0.8723	0.3538	-0.1608	0.9273
PE	No Target	0.0002	0.0004	0.0142	0.0000	2.4515	4.8033
	Vol Target	0.0002	0.0003	0.0026	0.0000	3.0096	4.3946
SF	No Target	0.0000	0.0001	0.8737	0.4176	0.1590	0.8106
	Vol Target	0.0000	0.0000	0.9672	0.7535	0.0411	0.3140

It is reasonable to conclude that the model provides a poor fit to the data. Most coefficients exhibit weak p-values and t-statistics, indicating limited statistical significance. Further work is required to improve the fit, potentially through alternative preprocessing or different model specifications (e.g., a logistic framework). Trading the raw OLS residuals yields weak Sharpe ratios; however, applying decile-based optimization and volatility targeting results in a modest improvement in portfolio performance.

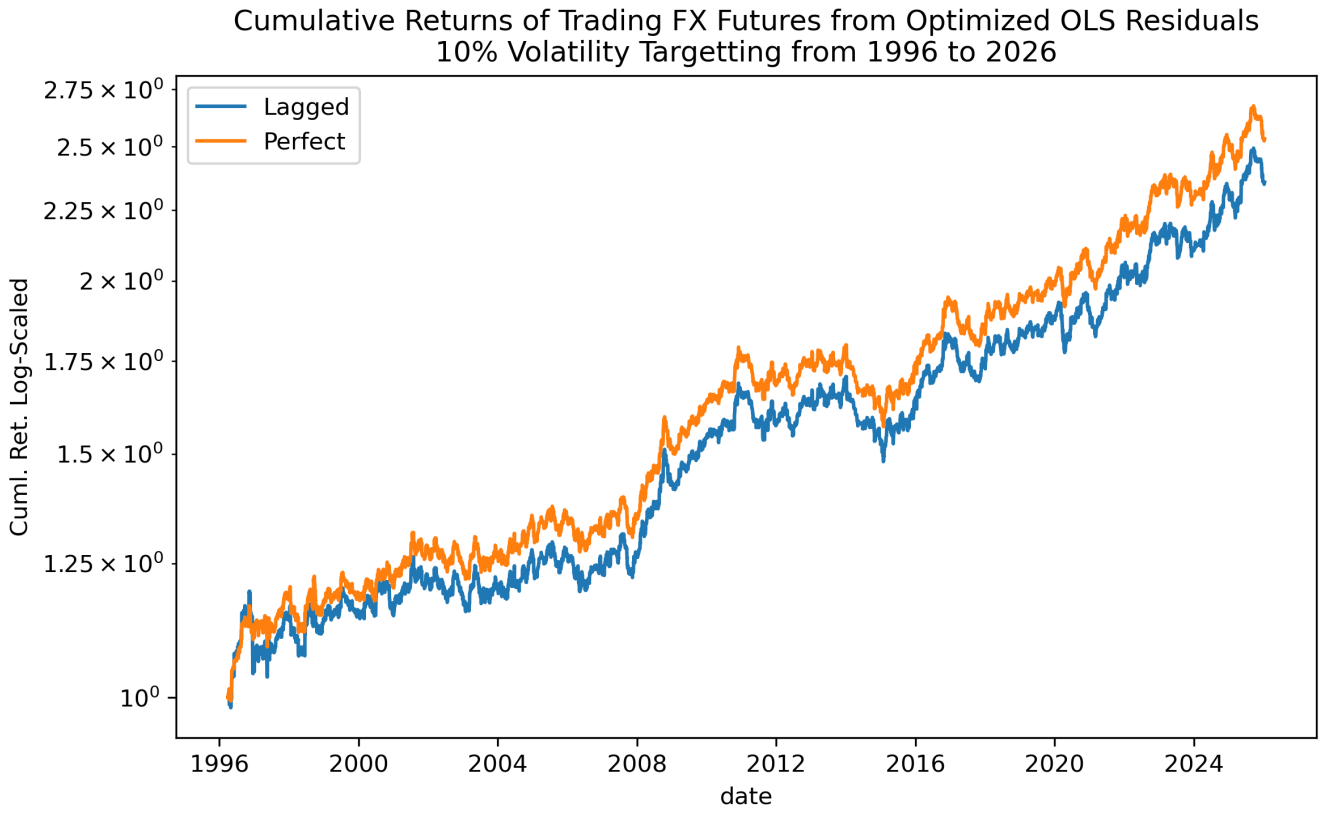
Below are the returns of trading the each optimized signal using the OLS returns



Below is a table of the sharpes of those strategies

Vol. Target	Future	AD	BP	BR	CD	EC	JY	NV	PE	SF
Lagged	Adj Sharpe	0.335	NaN	0.102	0.274	0.542	0.192	NaN	0.339	0.254
	Raw Sharpe	0.220	NaN	0.062	0.180	0.329	0.126	NaN	0.222	0.167
Perfect	Adj Sharpe	0.251	NaN	0.110	0.503	0.559	0.342	NaN	0.449	0.329
	Raw Sharpe	0.165	NaN	0.066	0.331	0.340	0.225	NaN	0.294	0.216

Taking an equal weight which is roughly equal volatility leads to the following portfolio



The sharpes of the portfolio are

group	Adj Sharpe	Raw Rtn
Lagged	0.548	0.502
Perfect	0.650	0.596

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